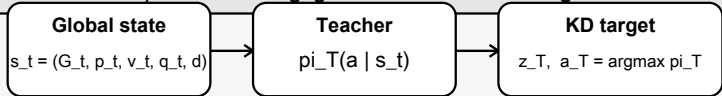
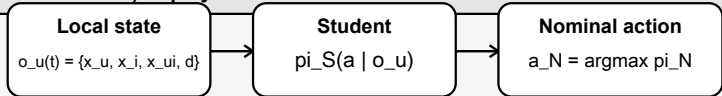


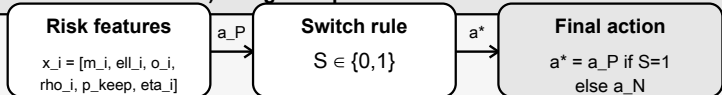
1) Offline training: global teacher $\rightarrow$ local targets



2) Deployment: local observation $\rightarrow$ nominal action



3) Safeguard: predictive risk switch



Equation view

Distillation

$$L_{\text{KD}} = T^2 D_{\text{KL}}(\pi_T || \pi_S) + \text{CE}(a_T, \pi_S) + L_{\text{oracle}}$$

Danger score

$$D_i = [g_m - m_i]_+ + [g_l - \ell_i]_+ + [g_o - o_i]_+ + [g_p - p_{\text{keep}}]_+$$

Safety gain

$$G(a_P, a_N) = Q(a_P) - Q(a_N)$$
$$Q_i = m_i + \ell_i + o_i + \rho_i + p_{\text{keep}}$$

Switch decision

$$S = 1[a_N = \text{DROP} \text{ or } D(a_N) > \tau \text{ or } G(a_P, a_N) > \delta]$$

Execution constraint

π_T is offline only; a^* uses local features.